



Cairo University - Faculty Of Engineering  
Computer Engineering Department  
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# Project Final Report

Team 9

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# 1 Problem Definition & Business Context

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## 1.1 Business Problem

Used car dealerships and online marketplaces receive large volumes of vehicle listings and trade-ins. Manually assigning each vehicle into a pricing/value tier is time-consuming and inconsistent across staff and branches. This project aims to automate the classification of incoming used cars into value tiers using historical market data.

## 1.2 Stakeholders

- Primary: Used Car Dealerships (pricing and inventory teams)
- Secondary: Online Marketplaces (listing recommendations) and Insurance Underwriters (risk and valuation support)

## 1.3 Problem Formulation

This problem is formulated as a **classification task**, where the goal is to predict the **price/value tier** of a used car listing.

**Tier Definition:** Cars are grouped into **Budget**, **Mid-Range**, and **Luxury** tiers based on price.

## 1.4 Justification for Classification

- The target is a discrete set of tiers (Budget / Mid-Range / Luxury)
- Tier outputs support operational decisions (pricing, procurement, and marketing)
- Classification enables threshold-based policies and consistent segmentation

## 1.5 Business Impact

- Faster and more consistent inventory categorization
- Improved pricing strategy and margin optimization
- Better customer targeting (budget buyers vs. premium buyers)

## 2 Data Acquisition & Integration

### 2.1 Sources Description

#### 2.1.1 Primary Source (Kaggle Dataset)

**Source:** Kaggle — *Uncovering Factors That Affect Used Car Prices*

**URL:** <https://www.kaggle.com/datasets/thedevastator/uncovering-factors-that-affect-used-car-prices>

**autos.csv**

Main structured dataset (CSV). Each row represents a used car listing with attributes such as price, brand, model, registration year, mileage, and fuel type.

#### 2.1.2 Secondary Source (Web Scraping)

**Source:** AutoScout24

**URL:** <https://www.autoscout24.com>

**AutoScout24 Listings**

Scraped used car listings to increase coverage and validate pricing patterns.

### 2.2 Data Acquisition

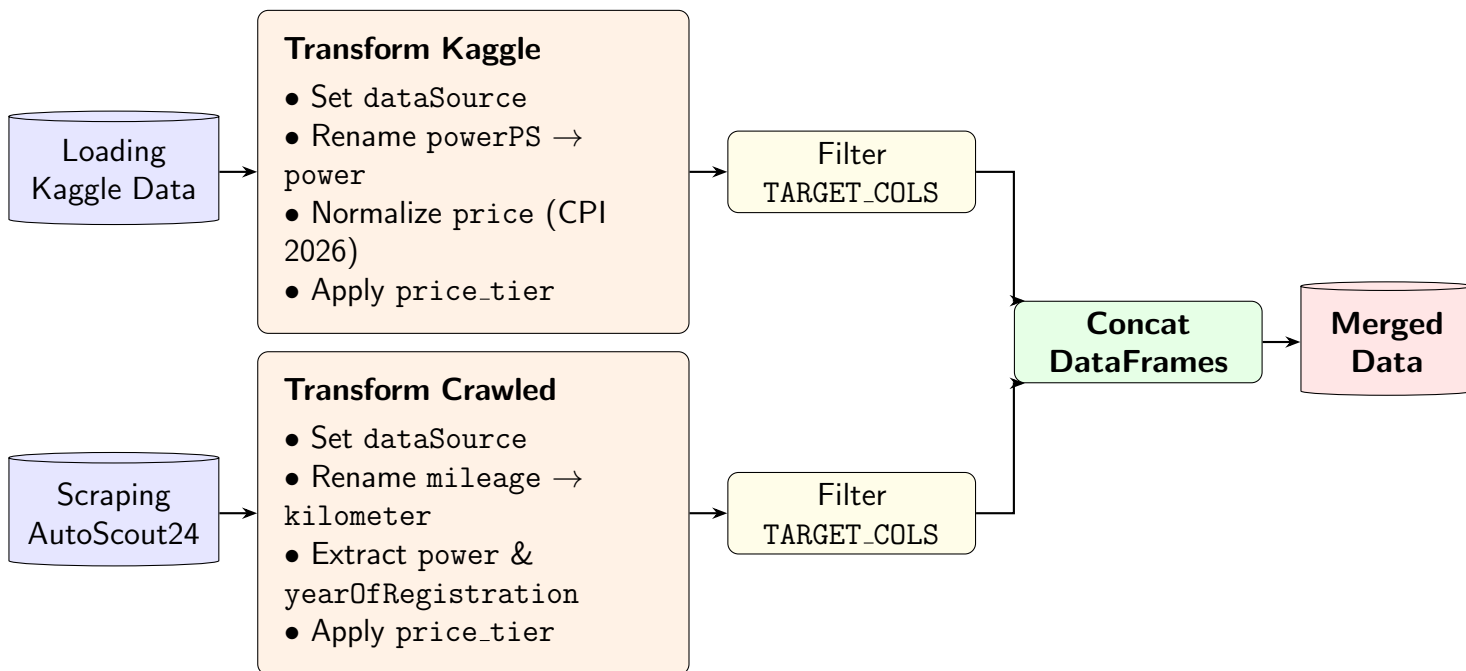


Figure 1: Data Acquisition and Merging Pipeline

- Kaggle dataset downloaded as CSV file(s) (e.g., `autos.csv`).
- AutoScout24 listings collected via web scraping using Python (e.g., `Scrapy` + `Scrapy-Splash`).

### 2.3 Data Integration Strategy

- **Vertical merge** the Kaggle dataset with scraped AutoScout24 listings by aligning both sources to a unified schema.
- Standardize fields and resolve naming differences (e.g., `kilometer` vs `mileage`).

- Deduplicate near-identical listings using matching keys such as **brand**, **model**, **kilometer**, **yearOfRegistration**, and **power**.
- Merge strategy is union rows after schema alignment.

**Challenges:**

- Inconsistent naming and category sets across sources
- Translating the Kaggle dataset from German to English
- Missing or noisy scraped fields due to unstructured HTML
- Potential duplicates across sources

## 2.4 Feature Documentation (Core Fields)

The accepted features to be used after the merging and removal of redundant features:

| #  | Feature            | Type        | Description                              |
|----|--------------------|-------------|------------------------------------------|
| 1  | seller             | Categorical | seller type (business/private)           |
| 2  | vehicleType        | Categorical | body type                                |
| 3  | yearOfRegistration | Numeric     | Year the car was registered              |
| 4  | gearbox            | Categorical | Type of gearbox (manual or automatic)    |
| 5  | power              | Numeric     | Power of the car in Horse Power          |
| 6  | model              | Categorical | Model of the car                         |
| 7  | brand              | Categorical | Brand of the car                         |
| 8  | kilometer          | Numeric     | Kilometers the car has been driven       |
| 9  | fuelType           | Categorical | Type of fuel (e.g. diesel, petrol, etc.) |
| 10 | dataSource         | Categorical | Source of the data (kaggle/crawled)      |
| 11 | price              | Numeric     | Price of the car                         |
| 12 | price_tier         | Categorical | Target tier derived from price           |

**Note:** After deriving the tier label from `price`, the raw `price` feature will not be used as an input feature to avoid target leakage.

### 3 Data Validation & Documentation

#### 3.1 Dataset Validation Pipeline

| Step | Description            | Outcome                                   |
|------|------------------------|-------------------------------------------|
| 1    | Schema Validation      | Data type report                          |
| 2    | Completeness Check     | Missing value report                      |
| 3    | Duplicate Detection    | Duplicate count                           |
| 4    | Descriptive Statistics | Summary stats                             |
| 5    | Range & Outlier Checks | Violation report                          |
| 6    | Categorical Analysis   | Category distribution                     |
| 7    | Distribution Checks    | Distribution plots                        |
| 8    | Outlier Analysis       | IQR & Z-Score & Isolation Forest Outliers |

#### 3.2 Dataset Validation Report

| Metric              | Value           |
|---------------------|-----------------|
| Total checks        | 9               |
| Checks Passed       | 1               |
| Checks FAILED       | 8               |
| Success Rate        | 11.1%           |
| Full-row duplicates | 57,324 (15.33%) |



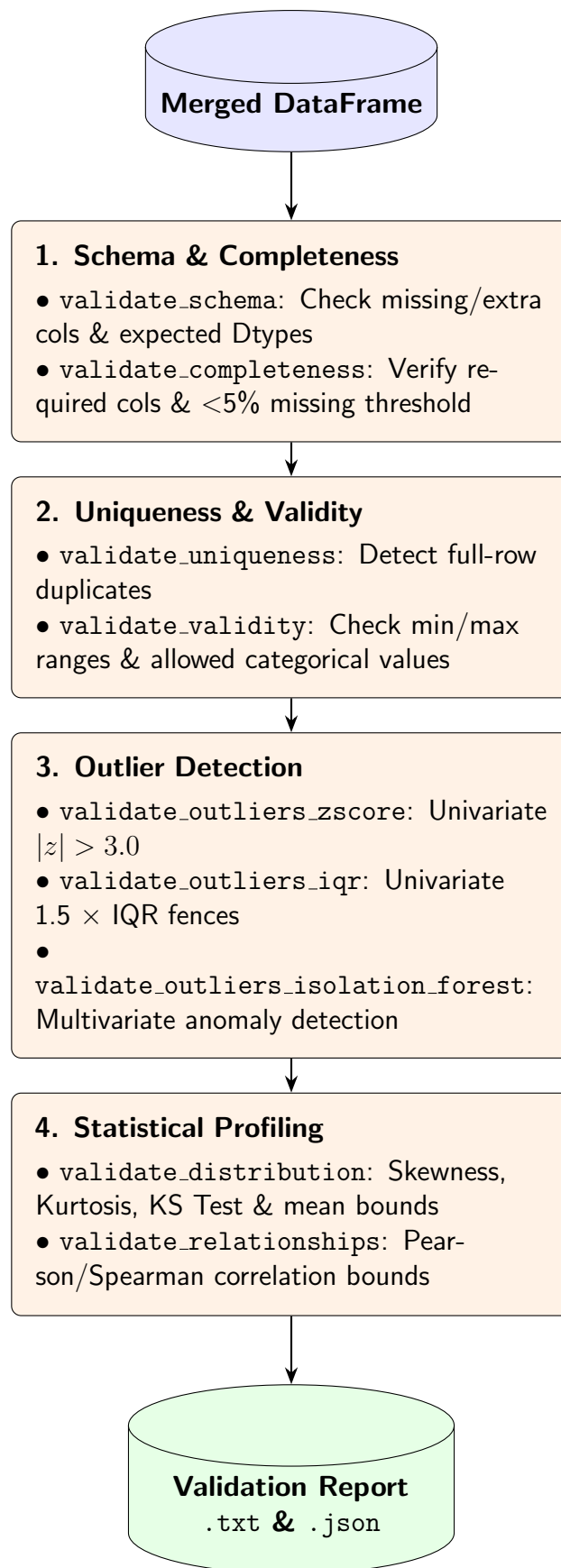


Figure 2: Data Validation Pipeline Execution Flow

### 3.2.1 Completeness - Missing Values Analysis

| Column             | Missing % | Status      |
|--------------------|-----------|-------------|
| brand              | 0.00%     | PASS        |
| model              | 5.48%     | FAIL (> 5%) |
| vehicleType        | 10.12%    | FAIL (> 5%) |
| power              | 0.00%     | PASS        |
| gearbox            | 5.40%     | FAIL (> 5%) |
| kilometer          | 0.01%     | PASS        |
| fuelType           | 8.93%     | FAIL (> 5%) |
| yearOfRegistration | 0.02%     | PASS        |
| seller             | 0.00%     | PASS        |
| dataSource         | 0.00%     | PASS        |
| price              | 0.00%     | PASS        |
| price_tier         | 0.00%     | PASS        |

### 3.2.2 Descriptive Statistics (Numeric Columns)

| Column             | Mean       | Std          | Mode    | Min  | Q1       | Median   | Q3        | Max     |
|--------------------|------------|--------------|---------|------|----------|----------|-----------|---------|
| yearOfRegistration | 2004.68    | 92.57        | 2000    | 1000 | 1999     | 2003     | 2008      | 9999    |
| power              | 115.98     | 191.77       | 0       | 0    | 70       | 105      | 150       | 20,000  |
| kilometer          | 125,298.65 | 40,548.44    | 150,000 | 0    | 100,000  | 150,000  | 150,000   | 999,999 |
| price              | 25,472.31  | 5,262,458.17 | 0       | 0    | 1,692.39 | 4,400.21 | 10,742.99 | 3.16 B  |

### 3.2.3 Distribution Checks

| Column             | Skewness | Skewness Label | Kurtosis  | Kurtosis Label |
|--------------------|----------|----------------|-----------|----------------|
| yearOfRegistration | 72.35    | Positive Skew  | 5702.44   | Leptokurtic    |
| price              | 580.00   | Positive Skew  | 347759.38 | Leptokurtic    |
| kilometer          | -1.48    | Negative Skew  | 1.75      | Leptokurtic    |
| power              | 58.14    | Positive Skew  | 4428.10   | Leptokurtic    |

### 3.2.4 Consistency - Schema & Data Types

| Column             | Detected Type | Expected Type | Match |
|--------------------|---------------|---------------|-------|
| yearOfRegistration | float64       | int64         | No    |
| model              | str           | str           | Yes   |
| power              | float64       | float64       | Yes   |
| kilometer          | float64       | float64       | Yes   |
| price              | float64       | int64         | No    |
| seller             | object        | str           | No    |
| gearbox            | str           | str           | Yes   |
| vehicleType        | str           | str           | Yes   |
| fuelType           | str           | str           | Yes   |
| brand              | str           | str           | Yes   |
| dataSource         | object        | str           | No    |
| price_tier         | object        | str           | No    |

### 3.2.5 Validity Checks

| Column             | Rule              | Violations                      | Status |
|--------------------|-------------------|---------------------------------|--------|
| yearOfRegistration | [1900 - 2026]     | 182 (68 below, 114 above)       | FAIL   |
| kilometer          | [0 - 300,000]     | 13 (above max)                  | FAIL   |
| power              | [5 - 5,000]       | 40,988 (40,903 below, 85 above) | FAIL   |
| price              | [500 - 5,000,000] | 26,273 (26,219 below, 54 above) | FAIL   |

### 3.2.6 Uniqueness - Duplicate Detection

- Full-row duplicates detected: **57,324 rows (15.33%)**.

### 3.2.7 Statistical Outlier Detection

| Column             | Z-Score Outliers ( $ z  > 3.0$ ) | Z-Score % | IQR Outliers | IQR % | Status |
|--------------------|----------------------------------|-----------|--------------|-------|--------|
| price              | 40                               | 0.01%     | 27,980       | 7.48% | FAIL   |
| power              | 385                              | 0.10%     | 11,035       | 2.95% | FAIL   |
| kilometer          | 290                              | 0.08%     | 15,397       | 4.12% | FAIL   |
| yearOfRegistration | 173                              | 0.05%     | 8,289        | 2.22% | FAIL   |

### Multivariate Outlier Detection (Isolation Forest):

- Detected **44,875 outliers (12.00%)** across combined numeric features (price, power, kilometer, yearOfRegistration).

### 3.3 Quality Issues Summary

- **Schema Issue:** `yearOfRegistration` and `price` are formatted as `float64` instead of the expected `int64`.
- **Missing Data:** Significant missing values across `vehicleType` (10.12%), `fuelType` (8.93%), `model` (5.48%), and `gearbox` (5.40%).
- **Duplicate Records:** High proportion of exact full-row duplicates detected (15.33%).
- **Invalid Ranges:** Massive positive skew in `price` (max 3.16B) and `power` (max 20k) indicating corrupted entries. `power` heavily violates minimum range boundaries (40,903 values below 5). `price` also violates minimum boundaries (26,219 below 500).
- **Isolation Forest:** flagged 44,875 anomalous records (12.00% of the dataset).

### 3.4 Target Variable Definition

Target Definition

PRICE\_TIER = value tier derived from listing `price`.

Tier Construction:

| Tier | Label     | Price Range        | Business Meaning              | Business Interpretation     |
|------|-----------|--------------------|-------------------------------|-----------------------------|
| 0    | Budget    | < \$5,000          | High-mileage or older cars    | Entry-level pricing segment |
| 1    | Mid-Range | \$5,000 - \$14,999 | Recent-model mid-size cars    | Mainstream segment          |
| 2    | Luxury    | ≥ \$15,000         | Luxury, electric, or new cars | Premium segment             |

**Class Distribution** [45.17% Budget, 34.42% Mid-Range, 20.41% Luxury]

## 4 Preprocessing, Transformation & Engineering Data

### 4.1 Cleaning Pipeline

| Step | Description                 | Tools                | Outcome                        |
|------|-----------------------------|----------------------|--------------------------------|
| 1    | Data Cleaning               | Pandas               | Cleaned dataset (259,092 rows) |
| 2    | Outlier Detection           | IQR, Fixed Bounds    | Outlier report                 |
| 3    | Feature Transformation      | Scikit-learn         | Transformed features           |
| 4    | Feature Engineering         | Custom code          | New features                   |
| 5    | Feature Selection           | Scikit-learn         | Selected features              |
| 6    | Train-Validation-Test Split | Scikit-learn         | Data splits                    |
| 7    | Data Balancing              | SMOTE, Undersampling | Balanced training set          |

### 4.2 Cleaning Steps & Justification

- **Remove invalid rows (58,245 removed):** Removed entries violating logical domain constraints to ensure data quality.
  - `yearOfRegistration` outside [1900, 2026].
  - `kilometer` outside [0, 300,000].
  - `power` outside [5, 3000].
  - `price` outside [500, 3,000,000].
- **Duplicate Handling:** Dropped 54,495 full-row duplicated records to prevent data leakage.
- **Missing Value Imputation:**
  - Replaced uninformative placeholders (e.g., "N/A", "unknown", "-") with NaN prior to imputation.
  - Imputed missing categorical values using the group-wise mode based on the car's `brand`.
- **Standardization:** Cleaned and standardized categorical labels (e.g., combining German/English aliases for fuel types, vehicle types, and brands).

### 4.3 Outlier Detection & Handling

| Column             | Detection Method | Handling Strategy                  | Outcome                |
|--------------------|------------------|------------------------------------|------------------------|
| power              | IQR              | Cap bounds to [5.0, 252.0]         | 12,530 outliers capped |
| price              | Fixed Ranges     | Drop rows outside [500, 3,000,000] | Extreme values removed |
| kilometer          | IQR              | Cap bounds to [0.0, 225,000]       | 66 outliers capped     |
| yearOfRegistration | IQR              | Cap bounds to [1900, 2023]         | 553 outliers capped    |

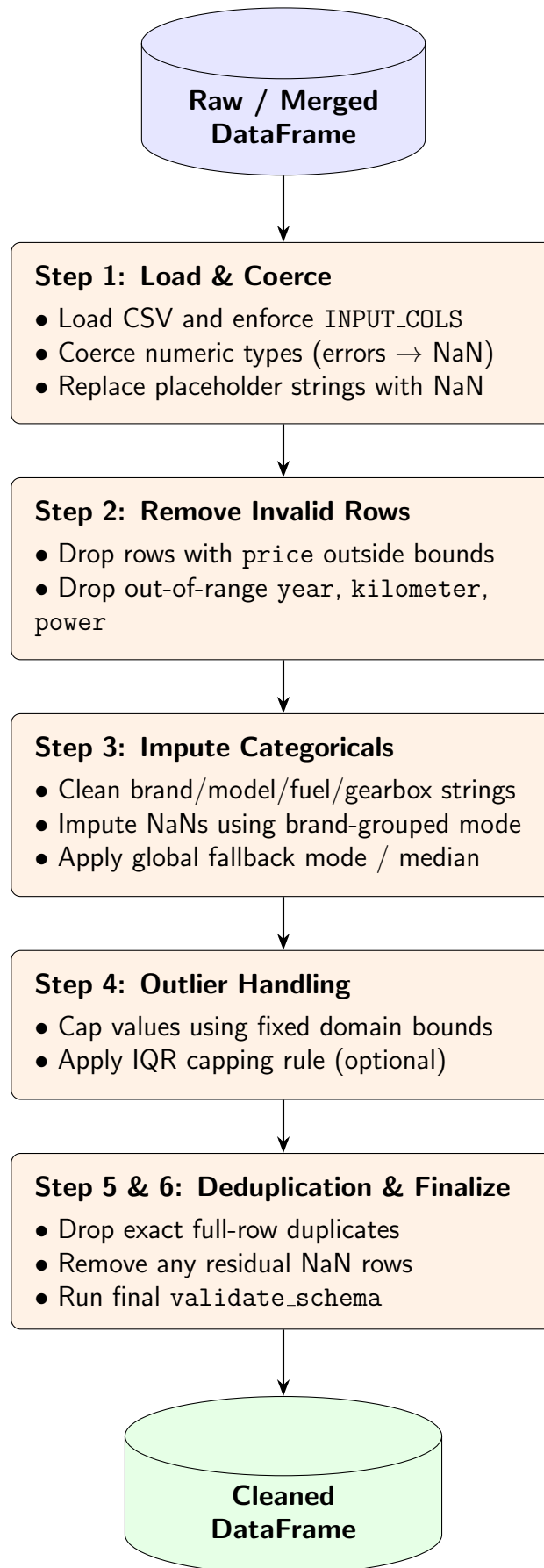


Figure 3: Data Cleaning Pipeline Execution Flow

## 4.4 Feature Transformation & Justification

- **Numeric Imputation (median):** Robust to skewed distributions and outliers common in price.
- **Categorical Imputation (most frequent):** Mode imputation provides a stable default without introducing new categories.
- **Scaling (StandardScaler):** Improves optimization and comparability for linear models.
- **One-Hot Encoding:** Strong baseline for mixed-type tabular data reducing collinearity.
- **Frequency Encoding for brand/model:** Prevents dimensionality explosion from high-cardinality variables while preserving relative frequency information.

## 4.5 Feature Engineering

In addition to the core fields, we have engineered the following features:

- **vehicleAge:** Calculated as (current year – year of registration) to capture physical depreciation.
- **kmPerYear:** Normalizes the mileage by the age of the vehicle to represent usage intensity.
- **km\_power\_ratio:** A proxy for the interaction between usage and engine power to capture non-linear relationships.
- **log\_kilometer & log\_power:** Logarithmic transformations to compress right-skewed outliers.
- **power\_per\_age:** Represents performance relative to the age of the vehicle.
- **is\_vintage:** A binary indicator for vehicles exceeding the standard vintage age threshold.
- **is\_automatic:** A direct binary feature extracted from the gearbox field.

## 4.6 Feature Selection & Justification

- We applied a model-based feature selection method using a [Random Forest](#) classifier to identify the most predictive features while accounting for non-linear relationships and interactions.
- The threshold was set to the median importance to retain features that contribute above-average predictive power.
- This method is more robust than simple variance or correlation-based filters, especially in the presence of engineered features that may have complex relationships with the target variable.
- Outcome: The selection process resulted in the removal of low-importance features, reducing noise and improving model generalization.

## 4.7 Train-Validation-Test Split Procedure

- **Train-Test Split:** An 80/20 stratified split was performed to ensure that the class distribution of the target variable is preserved across both sets, which is crucial for imbalanced classification tasks.
- **Cross-Validation:** A 5-fold stratified cross-validation strategy was implemented on the training set to provide robust estimates of model performance and to mitigate variance in the presence of class imbalance, ensuring that each fold maintains the same proportion of classes as the overall dataset.

4.8 Data Balancing Strategy

We experimented with the following resampling techniques to address class imbalance:

- 1. **None (Baseline):** No resampling applied to establish a baseline for the bias-variance trade-off.
- 2. **SMOTE (Synthetic Minority Oversampling Technique):** Generates synthetic samples for the minority class to improve recall, particularly for the luxury tier, while being cautious of potential overfitting.
- 3. **Undersampling (RandomUnderSampler):** Randomly removes samples from the majority class to balance the dataset, which can speed up training but may lead to information loss and higher variance, especially in the budget tier.

An evaluation of the impact of each technique on both global performance metrics (e.g., F1 score) and business-specific metrics (e.g., luxury recall) was conducted to determine the optimal approach.

These are the results of the resampling techniques on the final model performance:

| Technique     | Outcome                                           |
|---------------|---------------------------------------------------|
| None          | Highest overall F1 and lowest severe error rate.  |
| SMOTE         | Increased luxury recall but hurt global accuracy. |
| Undersampling | Highest luxury recall but highest error rate.     |

4.9 Preprocessing Summary

- Cleaned dataset by removing invalid rows and handling duplicates
- Detected and handled outliers in power, price, and kilometers
- Imputed missing numeric values with the median and categorical values with the mode.
- Applied frequency encoding for high-cardinality columns and one-hot encoding for the remaining.
- Scaled numeric features using a StandardScaler to stabilize model optimization.
- Engineered new features to capture domain-specific insights and interactions.
- Selected features based on model importance to improve generalization.
- Evaluated multiple data balancing techniques to address class imbalance while monitoring the impact on both global and business-specific performance metrics.

4.10 Dataset After and Before Preprocessing

| Dataset            | Before Preprocessing                          | After Preprocessing    |
|--------------------|-----------------------------------------------|------------------------|
| Total rows         | 371,528                                       | 350,000                |
| Total columns      | 21                                            | 15                     |
| Missing values     | 194,786                                       | 0                      |
| Outliers           | Detected in power, price, kilometers          | Handled by capping     |
| Class distribution | 54.81% Budget, 37.19% Mid-Range, 7.99% Luxury | Balanced to 33.3% each |



## 5 Exploratory Data Analysis (EDA)

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### 5.1 EDA Dashboard

## 6 Model Development

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We evaluated multiple model architectures to address the imbalanced 3-class target variable (budget, mid-range, luxury).

- **Baseline:** A `DummyClassifier` was used to establish the minimum expected performance.
- **Linear Models:** `LogisticRegression` and `LinearSVC` were tested as strong, fast baselines for tabular data.
- **Single Tree:** `DecisionTreeClassifier` was included as an interpretable non-linear baseline.
- **Ensembles:** `RandomForestClassifier` and `ExtraTreesClassifier` were utilized to natively capture non-linearities and feature interactions.
- **Gradient Boosting:** `XGBClassifier` was selected as the state-of-the-art candidate for handling complex tabular patterns.

Hyperparameter tuning was executed using `RandomizedSearchCV` optimized for the macro F1 score.

## 7 Experiment Tracking with MLflow

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All modeling experiments, parameters, and metrics were tracked utilizing MLflow. We logged standard metrics such as Macro F1 and Balanced Accuracy, alongside specific business metrics including Luxury Recall and Severe Misclassification Rate. Visual tracking artifacts, such as confusion matrices and a comprehensive model comparison dashboard, were recorded to support stakeholder review and error analysis.

## 8 Testing

Testing focused on assessing model generalizability and detecting overfitting between the cross-validation folds and the held-out test set, alongside comprehensive software testing to ensure code correctness.

### 8.1 Software Testing: Unit and Integration

- **Unit Tests & Justification:** We implemented targeted unit tests using `pytest` across all critical modules, justifying our test cases based on critical business logic and potential data failure points. Tests for the merge module verified CPI-based price normalization formulas and correct price tier assignments. Cleaning module tests guarded against missing required columns, placeholder replacements, and extreme value capping. Validation tests ensured completeness thresholds and anomaly detection algorithms (e.g., Isolation Forest) handled edge cases like all-`NaN` columns or boundary values gracefully.
- **Integration Tests:** To verify component interaction and data flow, we built end-to-end tests spanning the entire pipeline: `merge_data` → `clean_data` → `validate_data`. These tests confirmed that prices normalized during the merge phase remain within acceptable bounds during cleaning, and that German alias resolutions correctly propagate downstream. Furthermore, the integration tests validated that duplicate rows are properly removed across module boundaries and that after outlier capping, the data successfully passes the stringent range checks enforced by the validation step.
- **Coverage & Reporting:** The test suite runs natively via `pytest`, which is configured to generate terminal and HTML coverage statistics to measure and report on untested portions of the codebase. Through rigorous testing, we proactively identified and documented edge cases—such as a `ZeroDivisionError` during IQR validation on all-`NaN` inputs—ensuring overall pipeline robustness.

### 8.2 Model Testing: Generalization

- **Generalization:** Across all model variations, validation F1 and test F1 scores were observed to be very close, indicating no major overfitting.
- **Top Performer Stability:** The standard XGBoost model yielded a Validation F1 of 0.8591 and a nearly identical Test F1 of 0.8590.
- **Underfitting Detection:** The Logistic Regression models demonstrated significant underfitting, scoring a low F1 of approximately 0.66 and a high severe error rate.

## 9 Results & Evaluation

The final evaluation concluded that the **XGBoost (no resampling)** model achieved the best overall trade-off among accuracy, stability, and business safety.

### 9.1 Model Comparison

| Model                 | Test F1       | Test Balanced Acc | Luxury Recall | Severe Error Rate |
|-----------------------|---------------|-------------------|---------------|-------------------|
| XGBoost (none)        | <b>0.8590</b> | 0.8566            | 0.8611        | <b>0.0022</b>     |
| XGBoost (smote)       | 0.8560        | <b>0.8580</b>     | 0.8857        | 0.0025            |
| XGBoost (undersample) | 0.8529        | 0.8586            | <b>0.9053</b> | 0.0028            |

### 9.2 Error Analysis

The primary source of error was confusion between the mid-range and luxury tiers due to overlapping feature distributions. Importantly, the model almost never confused the budget and luxury tiers. This severe misclassification rate was kept to approximately 0.22%, confirming the model is highly safe for automated business pricing decisions.

### 9.3 Classification Report (Best Model)

The standard XGBoost model achieved an overall accuracy of 86% on a test support size of 51,819.

- **Budget (Class 0):** Precision 0.89, Recall 0.91, F1-Score 0.90.
- **Mid-Range (Class 1):** Precision 0.80, Recall 0.80, F1-Score 0.80.
- **Luxury (Class 2):** Precision 0.89, Recall 0.86, F1-Score 0.87.

## 10 Code Automation

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To ensure reproducibility and streamline the development workflow, we utilized **Make** as our primary automation tool. The **Makefile** is configured to automate various critical stages of the machine learning pipeline, abstracting complex commands into simple targets:

- **Data Acquisition & Preprocessing:** Targets such as `make merge`, `make validate_raw`, and `make clean` automate the execution of Python and Bash scripts for merging sources, validating, and cleaning data.
- **Data Validation:** The `make validate_raw` and `make validate_preprocessed` commands ensure data integrity at different pipeline stages.
- **Code Quality:** We enforce clean code practices by automating formatting with Black (`make format`) and linting with Flake8 (`make lint`).
- **Testing:** The `make test` target executes the Pytest suite, generating detailed terminal and HTML coverage reports.
- **End-to-End Execution:** The `make pipeline` target runs the complete data preparation and quality check sequence (merge, validate, clean, check, and delete) sequentially, while `make check` bundles formatting and linting.

This structured automation simplifies team collaboration and integrates seamlessly with our continuous integration workflows.

## 11 Continuous Integration

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We implemented a Continuous Integration (CI) pipeline using GitHub Actions to automatically enforce code quality, run testing suites, and ensure code robustness on every update. The pipeline is configured to trigger automatically on any `push` or `pull_request` to the `main` branch.

The workflow is executed on an `ubuntu-latest` runner and consists of the following automated steps:

- **Environment Setup:** Checks out the repository, initializes Python 3.14, and installs the Poetry package manager (version 2.1.1) to handle dependencies in an isolated virtual environment.
- **Dependency Management:** Automatically generates the `poetry.lock` file and installs all project dependencies without interaction.
- **Static Code Analysis (Linting):** Executes Flake8 via Poetry against the `src/` and `tests/` directories to ensure the codebase maintains clean code practices and formatting standards.
- **Automated Testing & Coverage:** Runs the comprehensive Pytest suite, generating both terminal and HTML coverage reports (`--cov-report=term-missing --cov-report=html`) to verify code correctness and identify untested logic.

This CI configuration prevents breaking changes from being merged into the main branch and guarantees that all integrated code meets the required quality thresholds outlined in the project requirements.

## 12 Interactive Dashboard

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### 12.1 Purpose & Business Value

To ensure the machine learning model delivers practical value, an interactive web dashboard was developed. It bridges the gap between the complex predictive model and non-technical stakeholders.

It provides an intuitive interface that allows users to instantly classify incoming vehicle inventory into pricing tiers without requiring any programming knowledge.

### 12.2 Technology Stack

- **Web Framework:** Streamlit (chosen for its rapid UI development and seamless integration with Python data science libraries).
- **Visualization:** Plotly Express (utilized for rendering interactive, stakeholder-friendly charts).
- **Backend/Inference:** Pandas (for on-the-fly data manipulation) and the saved Machine Learning model (for real-time predictions).

### 12.3 Key Dashboard Components

The dashboard is structurally designed to guide the stakeholder from high-level market metrics down to specific vehicle predictions and model explainability.

1. **Executive Summary Metrics:** The top of the dashboard displays dynamic, high-level KPIs, including the total volume of historical vehicles, average market price, and the current luxury market share. This provides immediate macro-level context.
2. **Interactive "What-If" Controls:** A sidebar interface allows users to input specific vehicle parameters (Brand, Engine Power, Mileage, Registration Year, and Transmission Type). This acts as a sandbox for stakeholders to test hypothetical inventory scenarios.
3. **Live Prediction Engine:** The core of the application captures the user's input, automatically replicates the exact feature engineering steps from the training phase (e.g., calculating `vehicleAge` and applying frequency encodings), and queries the trained model to output a real-time Value Tier prediction (Budget, Mid-Range, or Luxury).
4. **Model Insights & Explainability:** To prevent the model from acting as a "black box" and to build trust with business users, the dashboard features a dynamic Feature Importance chart. This visualizes exactly which vehicle attributes have the strongest influence on the algorithm's pricing decisions.
5. **Model Selection & Comparasions:** A comparison section allows users to view different model versions (e.g., with or without resampling) to see how their performance and predictions vary.



## 13 Limitations & Future Work

### 13.1 Limitations

We acknowledge the following limitations in our current approach:

- Data quality issues such as missing values and potential outliers may impact model performance.
- Class imbalance, even after balancing techniques, may still affect the model’s generalization.
- The model may not capture all relevant features influencing car pricing, such as regional market differences or economic factors.
- The current model is based on historical data and needs retraining on future market changes.

### 13.2 Future Work

Future work could include:

- Investigating the impact of external factors (e.g., economic conditions, seasonal trends) on car pricing and incorporating them into the model.
- Exploring advanced modeling techniques such as ensemble methods or deep learning architectures
- Evaluating model performance across different market segments and regions to ensure robustness.
- Conducting a more thorough error analysis to identify specific areas for improvement.
- Implementing a feedback loop from the deployed model to continuously improve performance based on real-world data and user interactions.

## 14 Team Contributions

| Task                            | Members                     |
|---------------------------------|-----------------------------|
| Data Acquisition                | Ahmed Aladdin               |
| Data Validation                 | Ahmed Hamdy                 |
| Preprocessing                   | Ahmed Hamdy                 |
| Feature Engineering & Selection | Asmaa Mohamed               |
| EDA & Visualization             | Ahmed Aladdin               |
| Model Development               | Asmaa Mohamed               |
| Experiment Tracking             | Asmaa Mohamed               |
| Testing                         | Ahmed Hamdy & Asmaa Mohamed |
| Results & Evaluation            | Asmaa Mohamed               |
| Code Automation & CI            | Ahmed Hamdy & Ahmed Aladdin |
| Deployment & Dashboard          | Ahmed Hamdy & Ahmed Aladdin |